# ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

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## C02-6

## EXPERIMENT 6

## Aim:

To write program for capacitance of parallel plate capacitor in three different dielectric medium with specific area, dielectric thickness and various dielectric constants using MATLAB.

### Test case:

Determination and plotting of capacitance of parallel plate capacitor for given area and dielectric thickness

### Program:

```
clc;
clear;
A=input("Enter the area value=");
 d=input("Enter the thickness value=");
epsilon0=8.854*10^-12; C=epsilon0*A/d;
 disp('The capacitance is');
 disp(C);
% Plot: capacitance versus plate separation for the given area dv = linspace(1e-3,10e-3,50);
Cv = epsilon0*A./dv;
figure; plot(dv*1e3,Cv,'LineWidth',2);
 grid on; xlabel('Plate separation d (mm)');
 ylabel('Capacitance (F)');
title('Parallel plate capacitance versus separation');
 % Validation using Laplace's equation (1-D finite-difference solution between the plates) V0 = 100;
N = 51;
V = zeros(1,N);
V(N) = V0;
for it = 1:20000 V(2:N-1) = 0.5*(V(1:N-2) + V(3:N));
% Laplace equation: d2V/dx2 = 0 end x = linspace(0,d,N);
E = -gradient(V,x);
 % E = -dV/dx C_lap = epsilon0*A*abs(mean(E))/V0;
```

% C = Q/V0 with Q = epsilon0*E*A fprintf('Capacitance from Laplace solution = %.4e F\n', C_lap);

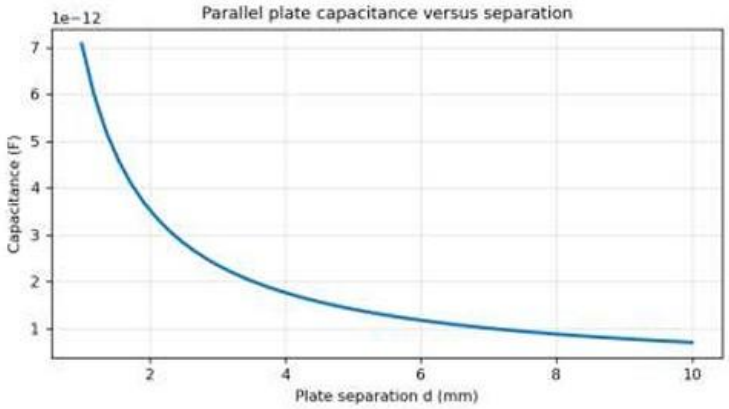
**Output:**

Enter the area value = 8e-4

Enter the thickness value = 3e-3

The capacitance is $2.36 \times 10^{-12}$ F

Capacitance from Laplace solution = 2.3611e-12 F




**Test case**

**(ii) Determination of capacitance of parallel plate capacitor with three different dielectric layers**

**Program:**

```
clc;
A=input("Enter the area value=");
d1=input("Enter thickness d1=");
d2=input("Enter thickness d2=");
d3=input("Enter thickness d3=");
er1=input("Enter dielectric constant er1=");
er2=input("Enter dielectric constant er2=");
er3=input("Enter dielectric constant er3=");
epsilon0=8.854e-12;
C=epsilon0*A/((d1/er1)+(d2/er2)+(d3/er3));
disp('The capacitance is');
disp(C);
```

**OUTPUT:**

Enter the area value = 8e-4

Enter thickness d1 = 1e-3

Enter thickness d2 = 3e-3

Enter thickness d3 = 2e-3

Enter dielectric constant er1 = 2.3

Enter dielectric constant er2 = 3.3

Enter dielectric constant er3 = 4

The capacitance is $3.84 \times 10^{-12}$ F

```
Enter the area value=8e-4
Enter thickness d1=1e-3
Enter thickness d2=3e-3
Enter thickness d3=2e-3
Enter dielectric constant er1=2.3
Enter dielectric constant er2=3.3
Enter dielectric constant er3=4
The capacitance is
   3.8415e-12
```

**Test case (iii)**

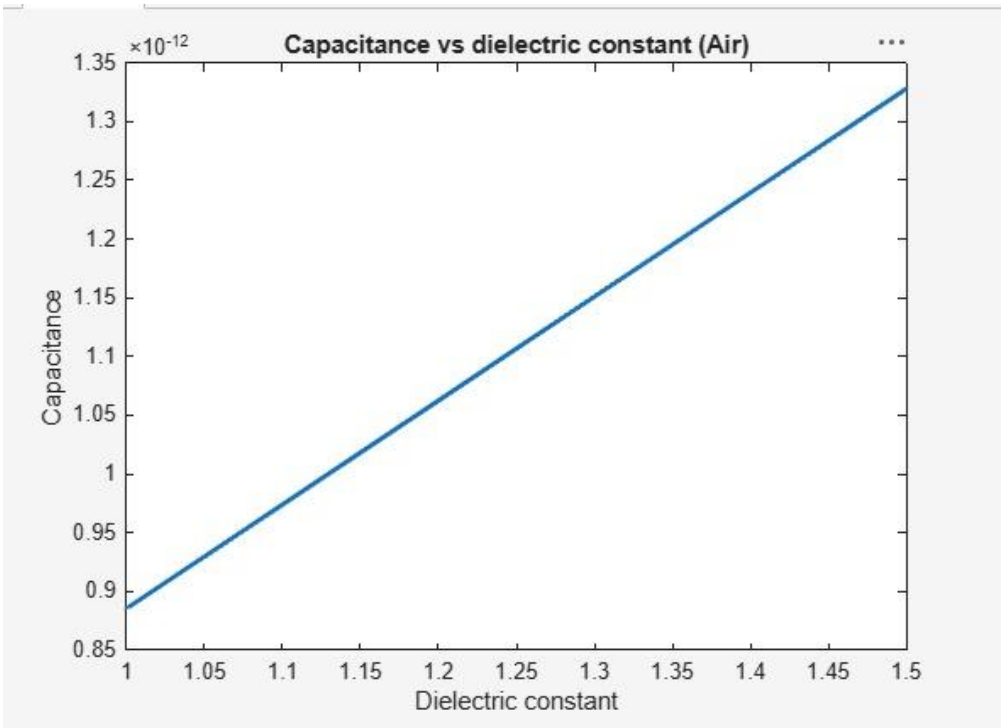
**Determination and plotting of capacitance variation with dielectric constant (Air)**

**Program:**

```
clc;
epsilon0 = 8.854e-12;
A = 1e-4; d = 1e-3;
er = linspace(1,1.5,100);
C = (epsilon0 .* er .* A) ./ d;
plot(er,C,'LineWidth',2);
xlabel('Dielectric constant');
ylabel('Capacitance');
title('Capacitance vs dielectric constant (Air)');
```

**Output:**

Capacitance increases linearly with dielectric constant



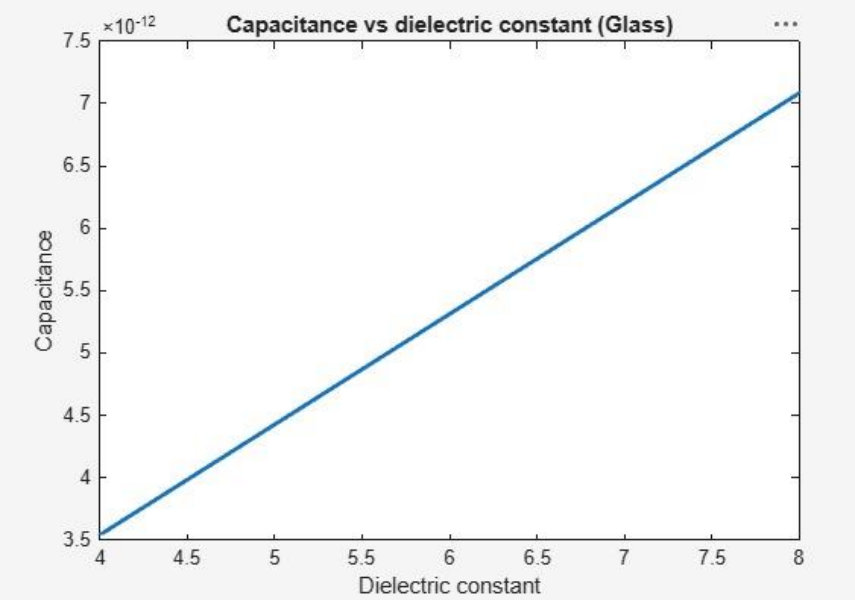

**Test case (iv)**

**Determination and plotting of capacitance variation with dielectric constant (Glass)**

```
clc;
epsilon0 = 8.854e-12;
A = 1e-4; d = 1e-3;
er = linspace(4,8,100);
C = (epsilon0 .* er .* A) ./ d;
plot(er,C,'LineWidth',2);
xlabel('Dielectric constant');
ylabel('Capacitance');
title('Capacitance vs dielectric constant (Glass)');
```

**Output:**

Higher dielectric → higher capacitance




**RESULTS:**

**Thus the program was verified for capacitance of parallel plate capacitor in different dielectric medium using MATLAB successfully.**

| S. No. | Objective | Assessment Criterion | Excellent – 4 | Good – 3 | Satisfactory – 2 | Needs Improvement – 1 | Weighting (%) | Marks given | Marks obtained |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Set capacitor geometry and material properties | Set capacitor geometry and material properties | Correctly enters plate area, separation and relative permittivity with SI units. | Inputs are correct with a minor unit/ property issue. | Several inputs need correction. | Geometry or dielectric inputs are invalid. | 25 | 2.5 | |
| 2 | Calculate capacitance in MATLAB | Calculate capacitance in MATLAB | Uses the correct capacitance relation and calculates values for all specified media. | Calculations are mostly correct with minor errors. | Only some cases are calculated correctly. | Capacitance calculations are incorrect. | 25 | 2.5 | |
| 3 | Compare dielectric and geometry effects | Compare dielectric and geometry effects | Correctly evaluates trends versus permittivity/area/ separation and validates the model. | Trend and validation are mostly correct with minor omissions. | Trend discussion or validation is incomplete. | Incorrectly interprets parameter effects. | 30 | 3 | |
| 4 | Document engineering observations | Document engineering observations | Provides a complete table/plot with units and | Documentation and explanation | Documentation and explanation are | Results are incomplete or unexplained. | 20 | 2 | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | explains dielectric influence. | are mostly complete. | mostly complete. | | | | |
| Total | | | | | | | 100 | 10 | |